



BOOLEAN ALGEBRA

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BOOLEAN ALGEBRA

- Boolean constants and variables are allowed to have only two possible values, 0 or 1.
- A Boolean variable is a quantity that may, at different times, be equal to either 0 or 1.
- Boolean variables are often used to represent the voltage level present on a wire or at the input/output terminals of a circuit.

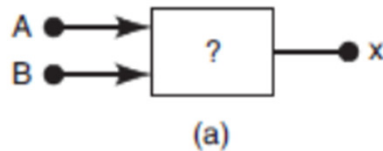
Logic 0	Logic 1
False	True
Off	On
LOW	HIGH
No	Yes
Open switch	Closed switch

TRUTH TABLE

A truth table is a means for describing how a logic circuit's output depends on the logic levels present at the circuit's inputs.

Diagram (a) shows a logic circuit with two inputs, A and B, entering a box labeled with a question mark. The output of the box is labeled x. Above the circuit, a truth table is shown with columns for Inputs (A, B) and Output (x). The inputs are labeled with arrows pointing to the circuit inputs, and the output is labeled with an arrow pointing to the circuit output.

Inputs		Output
A	B	x
0	0	1
0	1	0
1	0	1
1	1	0



A	B	C	x
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	1

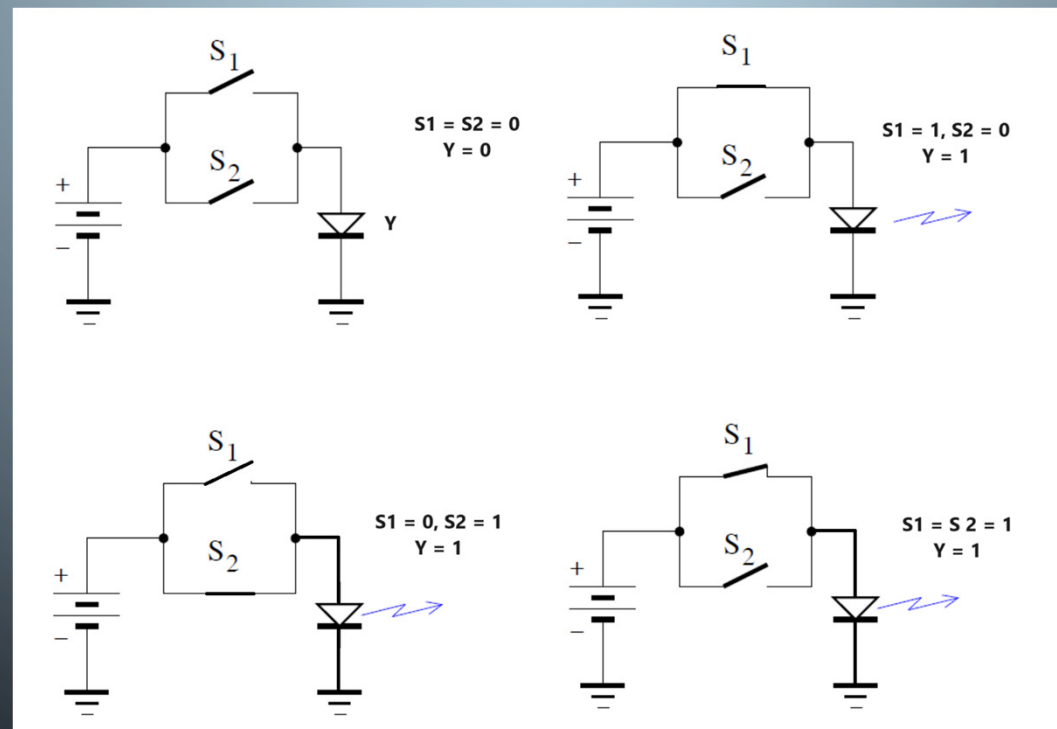
(b)

A	B	C	D	x
0	0	0	0	0
0	0	0	1	0
0	0	1	0	0
0	0	1	1	1
0	1	0	0	1
0	1	0	1	0
0	1	1	0	0
0	1	1	1	1
1	0	0	0	0
1	0	0	1	0
1	0	1	0	0
1	0	1	1	1
1	1	0	0	0
1	1	0	1	0
1	1	1	0	0
1	1	1	1	1

(c)

OR LOGICAL GATE – EQUIVALENT ELECTRICAL CIRCUIT

The LED lights up when either or both switches, S1 and S2, are closed, indicating a complete circuit.

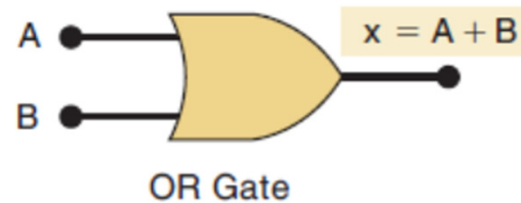


OR LOGICAL GATE – TRUTH TABLE AND SYMBOL

OR

A	B	$x = A + B$
0	0	0
0	1	1
1	0	1
1	1	1

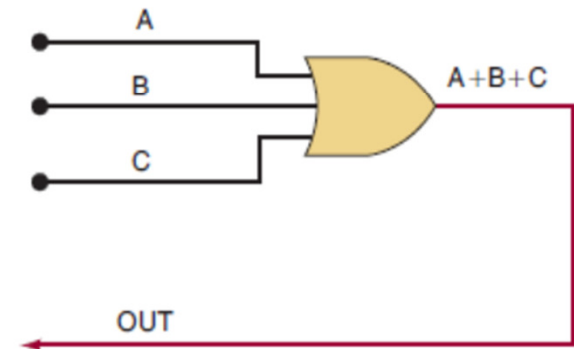
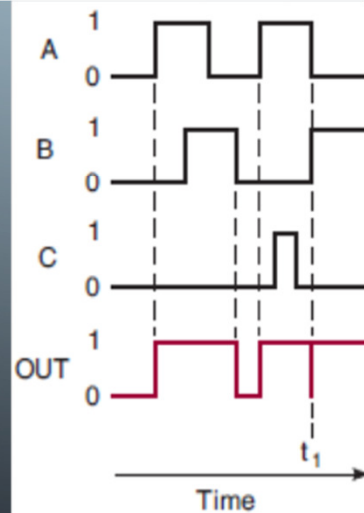
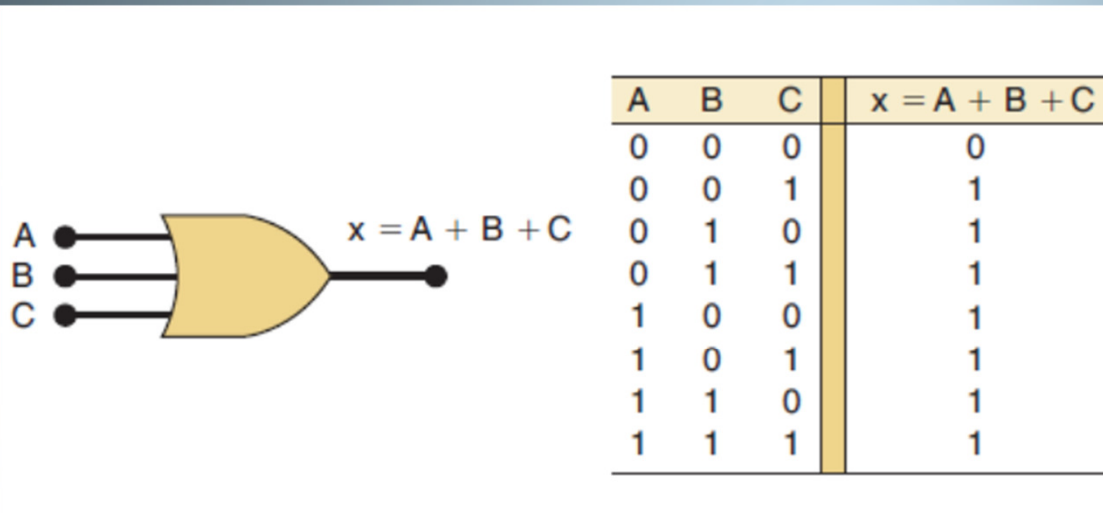
(a)



(b)

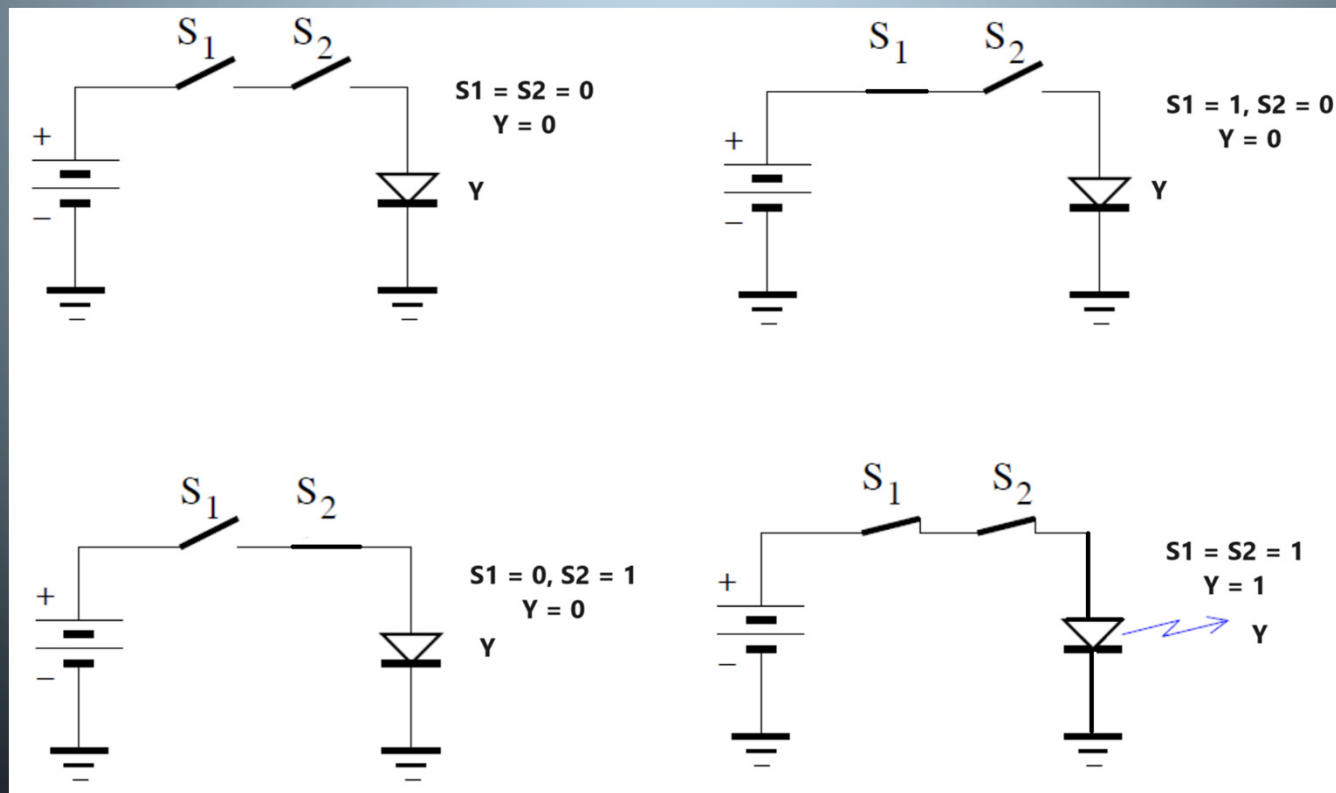
Switch		Output	Remarks
S2	S1	Y	
Open	Open	OFF	LED is OFF
Open	Closed	ON	LED is ON
Closed	Open	ON	LED is ON
Closed	Closed	ON	LED is ON

OR LOGICAL GATE – TRUTH TABLE AND SYMBOL



AND LOGICAL GATE – EQUIVALENT ELECTRICAL CIRCUIT

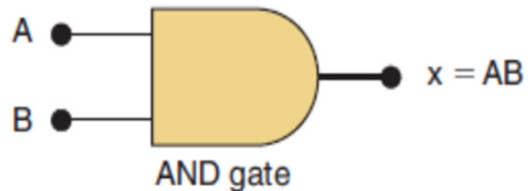
The LED lights up only when both switches, S_1 and S_2 , are closed, indicating a complete circuit.



AND LOGICAL GATE – TRUTH AND SYMBOL

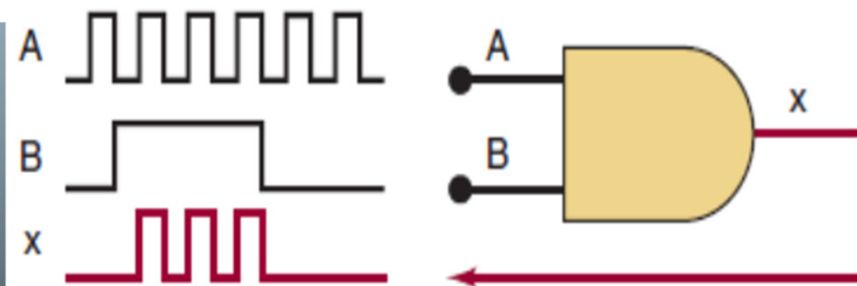
AND		
A	B	$x = A \cdot B$
0	0	0
0	1	0
1	0	0
1	1	1

(a)



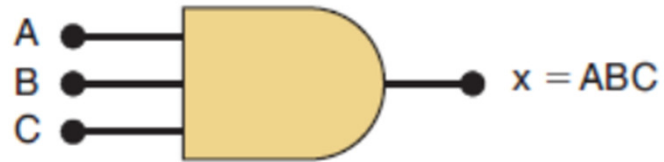
(b)

Switch		Output	Remarks
S2	S1	Y	
Open	Open	OFF	LED is OFF
Open	Closed	OFF	LED is OFF
Closed	Open	OFF	LED is OFF
Closed	Closed	ON	LED is ON

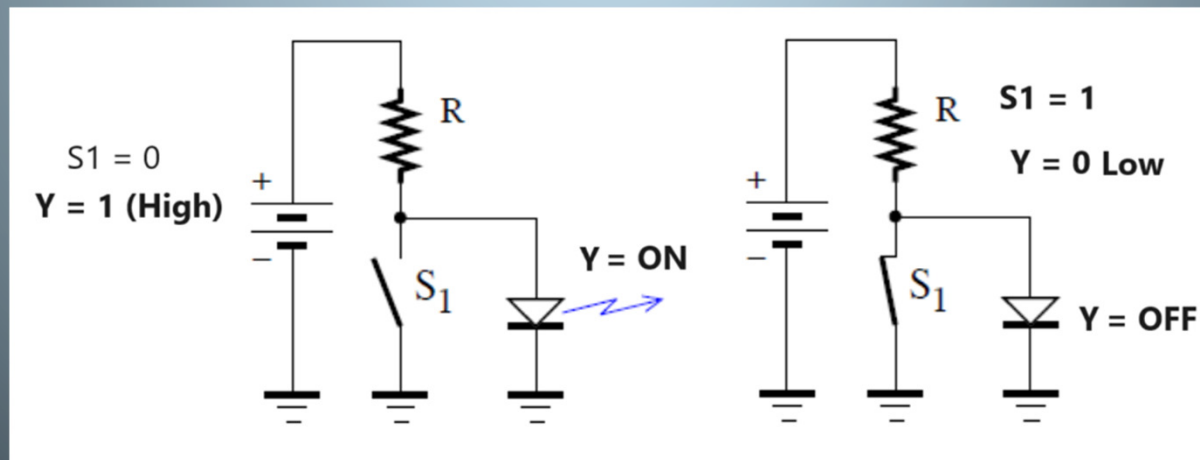


AND LOGICAL GATE – TRUTH AND SYMBOL

A	B	C	$x = ABC$
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	1



NOT LOGICAL GATE – ELECTRICAL EQUIVALENT CIRCUIT



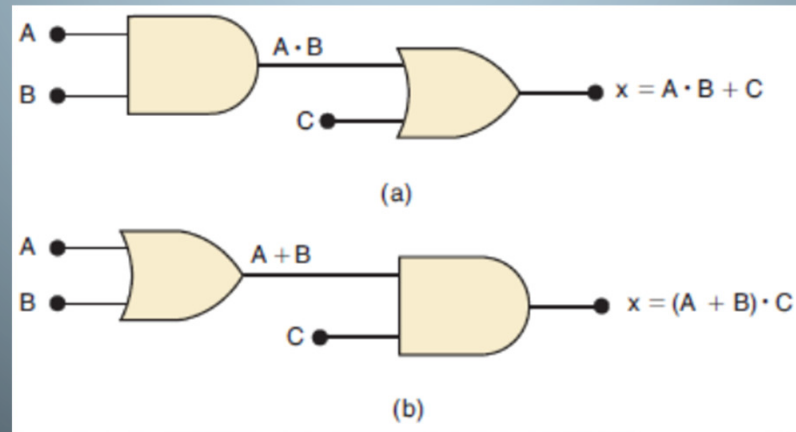
Switch	Output	Remarks
S_1	Y	
Open	ON	LED glows
Closed	OFF	LED is off

OPERATOR PRECEDENCE

(1) The expression $A \cdot B + C$ can be interpreted in two different ways: (1) $A \cdot B$ is ORed with C .

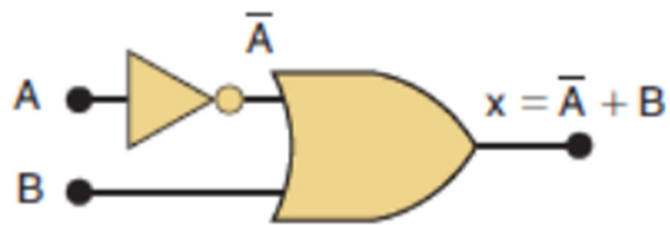
or

(2) A is ANDed with the term $B + C$.

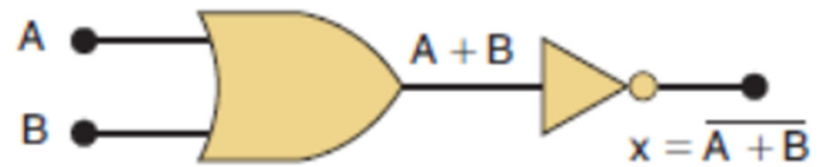


If an expression contains both AND & OR operations, the **AND operations are performed first**, unless there **are parentheses** in the expression, in which case the operation inside the parentheses is to be performed first.

CIRCUITS CONTAINING INVERTERS

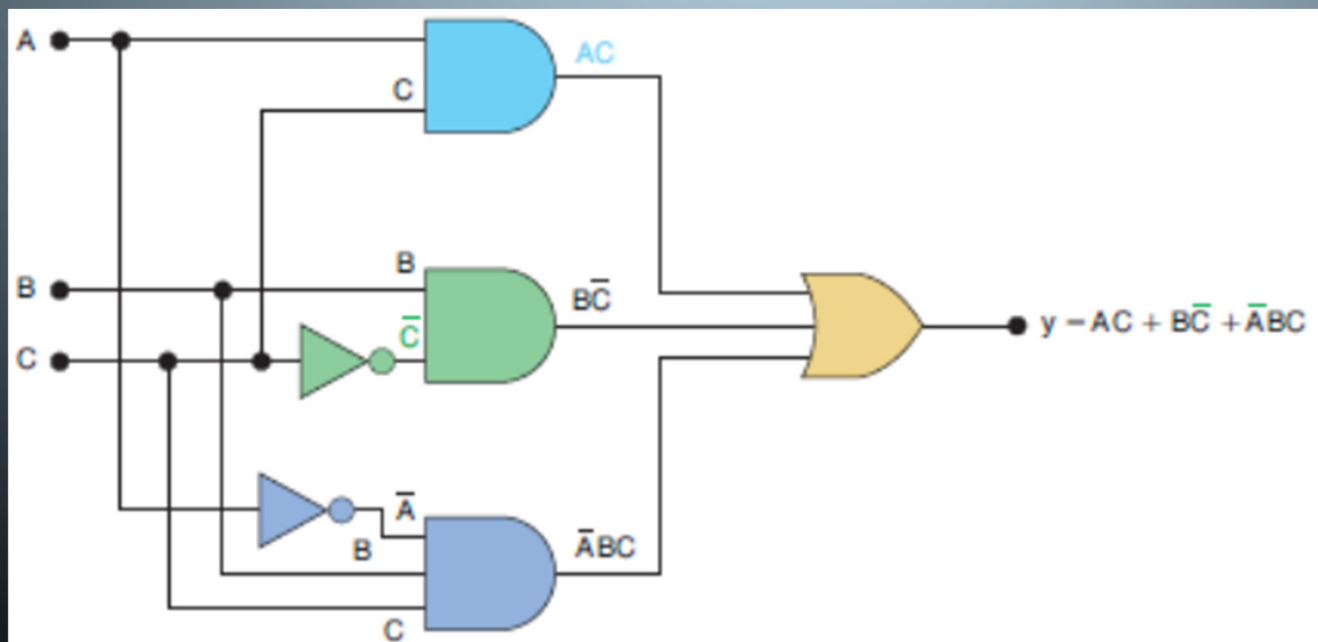
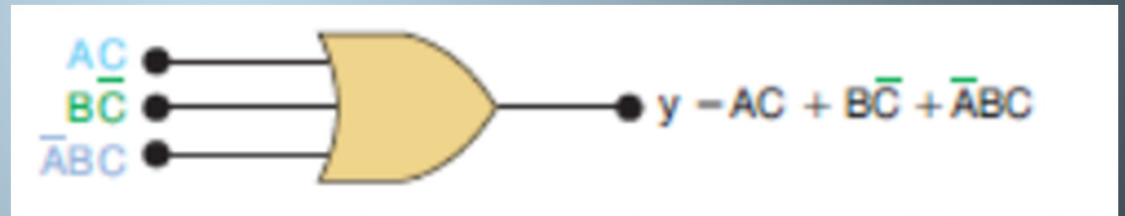


(a)

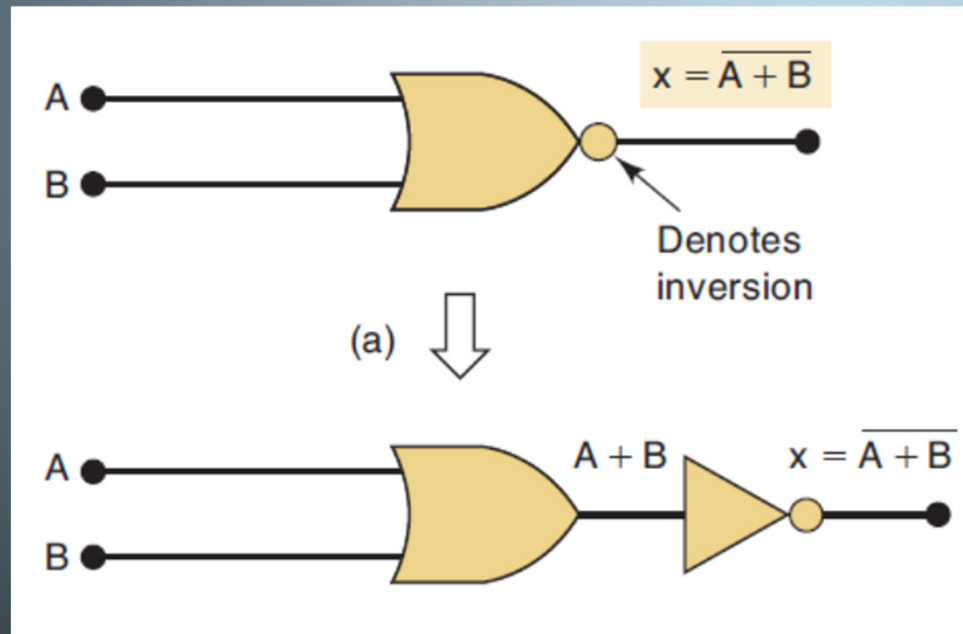


(b)

CIRCUITS FROM BOOLEAN ALGEBRA

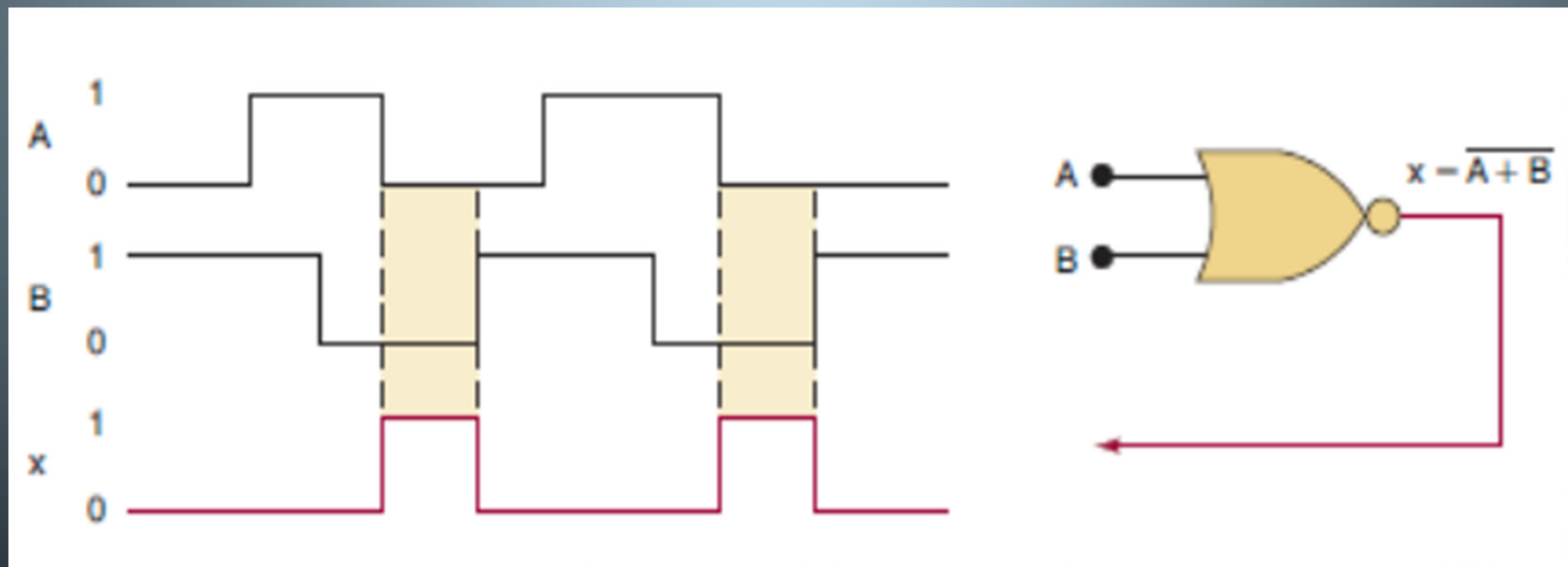


NOR LOGIC GATES

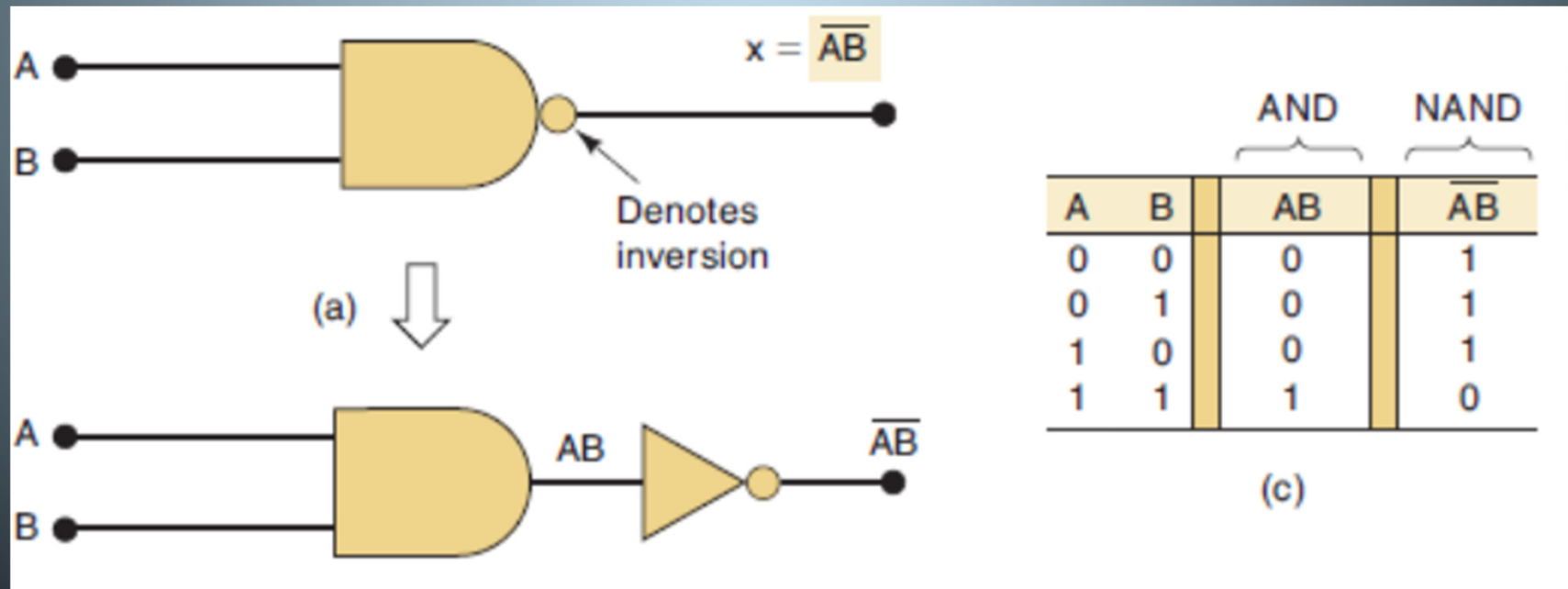


		OR		NOR	
		$A + B$		$\overline{A + B}$	
A	B	$A + B$		$\overline{A + B}$	
0	0	0		1	
0	1	1		0	
1	0	1		0	
1	1	1		0	

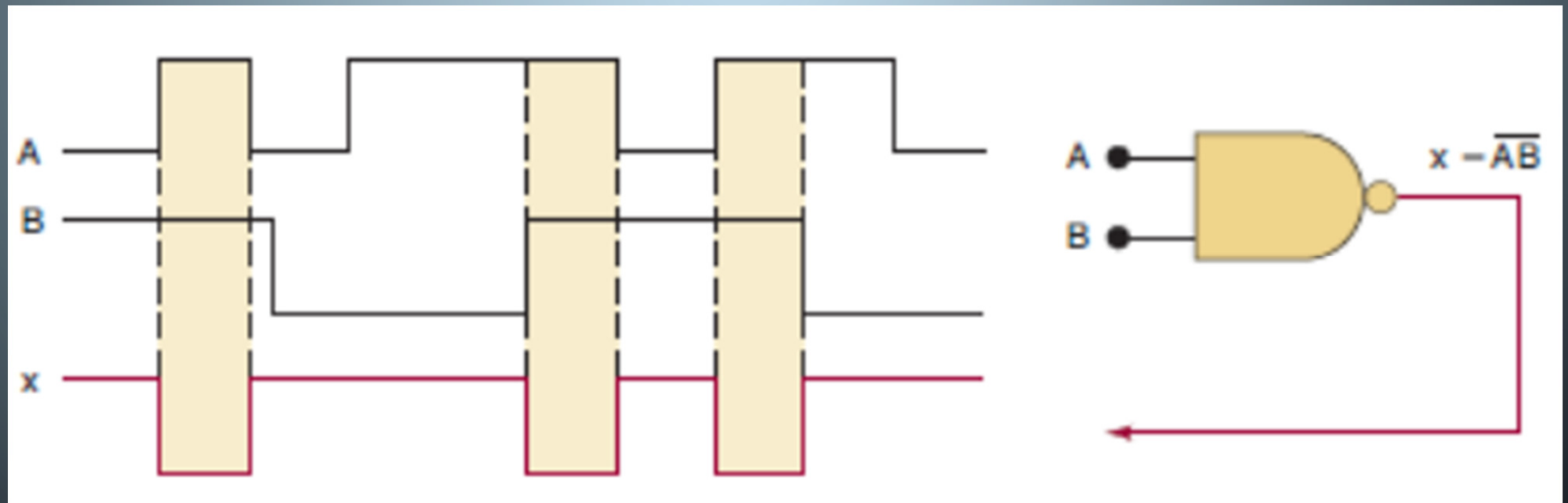
NOR LOGIC GATES – TIMING DIAGRAM



NAND LOGIC GATES



NAND LOGIC GATES-TIMING DIAGRAM



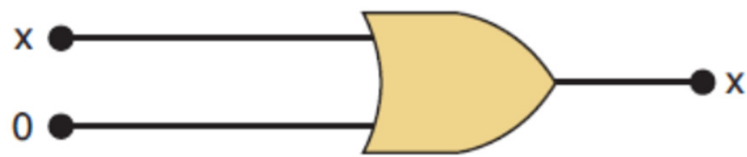
BOOLEAN ALGEBRA-ADDITION

$$0 + 0 = 0$$

$$0 + 1 = 1$$

$$1 + 0 = 1$$

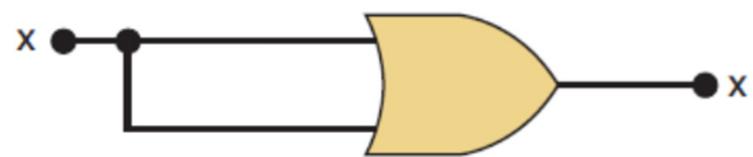
$$1 + 1 = 1$$



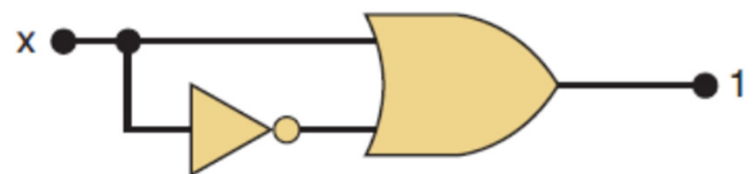
$$x + 0 = x$$



$$x + 1 = 1$$



$$x + x = x$$



$$x + \bar{x} = 1$$

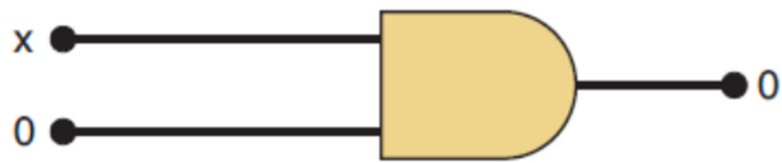
BOOLEAN ALGEBRA-MULTIPLICATION

$$0 \cdot 0 = 0$$

$$0 \cdot 1 = 0$$

$$1 \cdot 0 = 0$$

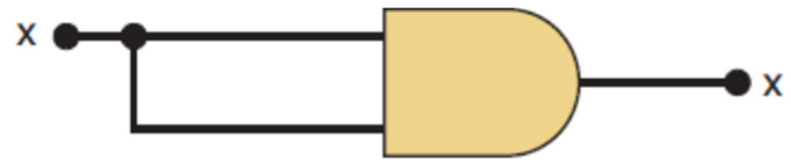
$$1 \cdot 1 = 1$$



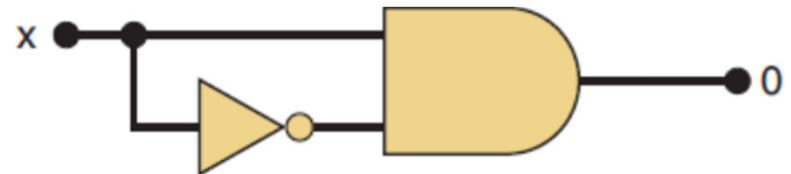
$$x \cdot 0 = 0$$



$$x \cdot 1 = x$$



$$x \cdot x = x$$



$$x \cdot \bar{x} = 0$$

BOOLEAN ALGEBRA- SINGLE VARIABLE THEOREMS

Annulment law

A variable **AND**ed with 0 gives 0, while a variable **OR**ed with 1 gives 1, i.e.,

$$\underline{A} \cdot 0 = 0$$

$$A + 1 = 1$$

Identity law

In this law variable remains unchanged it is ORed with '0' or ANDed with '1', i.e.,

$$A \cdot 1 = A$$

$$A + 0 = A$$

Idempotent law

A variable remains unchanged when it is ORed or ANDed with itself, i.e.,

$$A + A = A$$

$$\underline{A} \cdot A = A$$

Complement law

In this Law if a complement is added to a variable, it gives one, if a variable is multiplied with its complement, it results in '0', i.e.,

$$A + \bar{A} = 1$$

$$\underline{A} \cdot \bar{A} = 0$$

Double Negation Law

A variable with two negations, its symbol gets cancelled out and original variable is obtained, i.e.,

$$\overline{\bar{A}} = A$$

BOOLEAN ALGEBRA-COMMUTATIVE PROPERTY

Commutative property Boolean addition is commutative, given by

$$A + B = B + A \quad (1a)$$

According to this property, the order of the OR operation conducted on the variables makes no difference. Boolean algebra is also commutative over multiplication, given by

$$A \cdot B = B \cdot A \quad (1b)$$

This means that the order of the AND operation conducted on the variables makes no difference.

BOOLEAN ALGEBRA-ASSOCIATIVE PROPERTY

Associative property The associative property of addition is given by

$$A + (B + C) = (A + B) + C \quad (2a)$$

The OR operation of several variables results in the same, regardless of the grouping of the variables. The associative law of multiplication is given by

$$A \cdot (B \cdot C) = (A \cdot B) \cdot C \quad (2b)$$

According to this law, it makes no difference in what order the variables are grouped during the AND operation of several variables.

BOOLEAN ALGEBRA-DISTRIBUTIVE PROPERTY

$$A + BC = (A + B)(A + C)$$

$$\begin{aligned} A + BC &= A \cdot 1 + BC && (\because A \cdot 1 = A) \\ &= A(1 + B) + BC && (\because 1 + B = 1) \\ &= A \cdot 1 + AB + BC && (\because A(B + C) = AB + BC) \\ &= A \cdot (1 + C) + AB + BC && (\because 1 + C = 1) \\ &= A \cdot 1 + AC + AB + BC \\ &= A \cdot A + AC + AB + BC && (\because A \cdot A = A) \\ &= A(A + C) + B(A + C) \\ &= (A + B)(A + C) \end{aligned}$$

Similarly, we can proof the following expression

$$A \cdot (B + C) = A \cdot B + A \cdot C$$

BOOLEAN ALGEBRA - **ABSORPTION LAW**

$$\boxed{A + AB = A}$$

Proof

$$\begin{aligned} A + AB &= A \cdot 1 + AB \\ &= A(1 + B) \\ &= A \cdot 1 = A \end{aligned}$$

BOOLEAN ALGEBRA - **ABSORPTION LAW**

$$\boxed{A \cdot (A + B) = A}$$

Proof

$$\begin{aligned} A(A + B) &= A \cdot A + A \cdot B \\ &= A + AB \\ &= A(1 + B) \\ &= A \cdot 1 \\ &= A \end{aligned}$$

BOOLEAN ALGEBRA - **ABSORPTION LAW**

$$\boxed{A + \bar{A}B = A + B}$$

Proof

$$\begin{aligned} A + \bar{A}B &= (A + \bar{A})(A + B) \quad [\because A + BC = (A + B)(A + C)] \\ &= 1 \cdot (A + B) \quad (\because A + \bar{A} = 1) \\ &= A + B \end{aligned}$$

BOOLEAN ALGEBRA - ABSORPTION LAW

$$\boxed{A \cdot (\bar{A} + B) = AB}$$

Proof $A \cdot (\bar{A} + B) = A \cdot \bar{A} + AB$
 $= AB (\because A\bar{A} = 0)$

BOOLEAN ALGEBRA – CONSENSUS LAW

$$AB + \bar{A}C + BC = AB + \bar{A}C$$

Proof

$$\begin{aligned} AB + \bar{A}C + BC &= AB + \bar{A}C + BC \cdot 1 \\ &= AB + \bar{A}C + BC(A + \bar{A}) && (\because A + \bar{A} = 1) \\ &= AB + \bar{A}C + ABC + \bar{A}BC \\ &= AB(1 + C) + \bar{A}C(1 + B) && (\because 1 + B = 1 = 1 + C) \\ &= AB + \bar{A}C \end{aligned}$$

$$F = AB + B\bar{C} + AC$$

Redundant term

Complemented and
Uncomplemented variable

BOOLEAN ALGEBRA – CONSENSUS LAW

$$(A + B)(\bar{A} + C)(B + C) = (A + B)(\bar{A} + C)$$

$$\begin{aligned}(A + B)(\bar{A} + C)(B + C) &= (A + B)(\bar{A} + C)(B + C + 0) \\&= (A + B)(\bar{A} + C)(B + C + A\bar{A}) \\&= (A + B)(\bar{A} + C)(B + C + A)(B + C + \bar{A}) \\&\quad [\because A + BC = (A + B)(A + C)] \\&= (A + B)(A + B + C)(\bar{A} + C)(\bar{A} + C + B) \\&= (A + B)(\bar{A} + C) \\&\quad [\because A(A + B) = A]\end{aligned}$$

$$F = (A+B). (\bar{A}+C). (B+C)$$

Normal and Complemented term

Redundant Term

BOOLEAN ALGEBRA – PRINCIPLE OF DUALITY

The principle of duality is a fundamental concept in Boolean algebra and other areas of mathematics.

It states that every algebraic expression remains valid if we interchange the AND & OR operators and swap 0's with 1's and vice versa.

Steps to Apply the Duality Principle in Boolean Algebra:

- Change each AND operation to an OR operation.
- Change each OR operation to an AND operation.
- Replace 0 with 1.
- Replace 1 with 0.

Example:

Original Expression: $(A \cdot 0 = 0)$

Dual Expression: $(A + 1 = 1)$

BOOLEAN ALGEBRA – DE-MORGAN'S THEOREM

De-Morgan's theorem

The two theorems are:

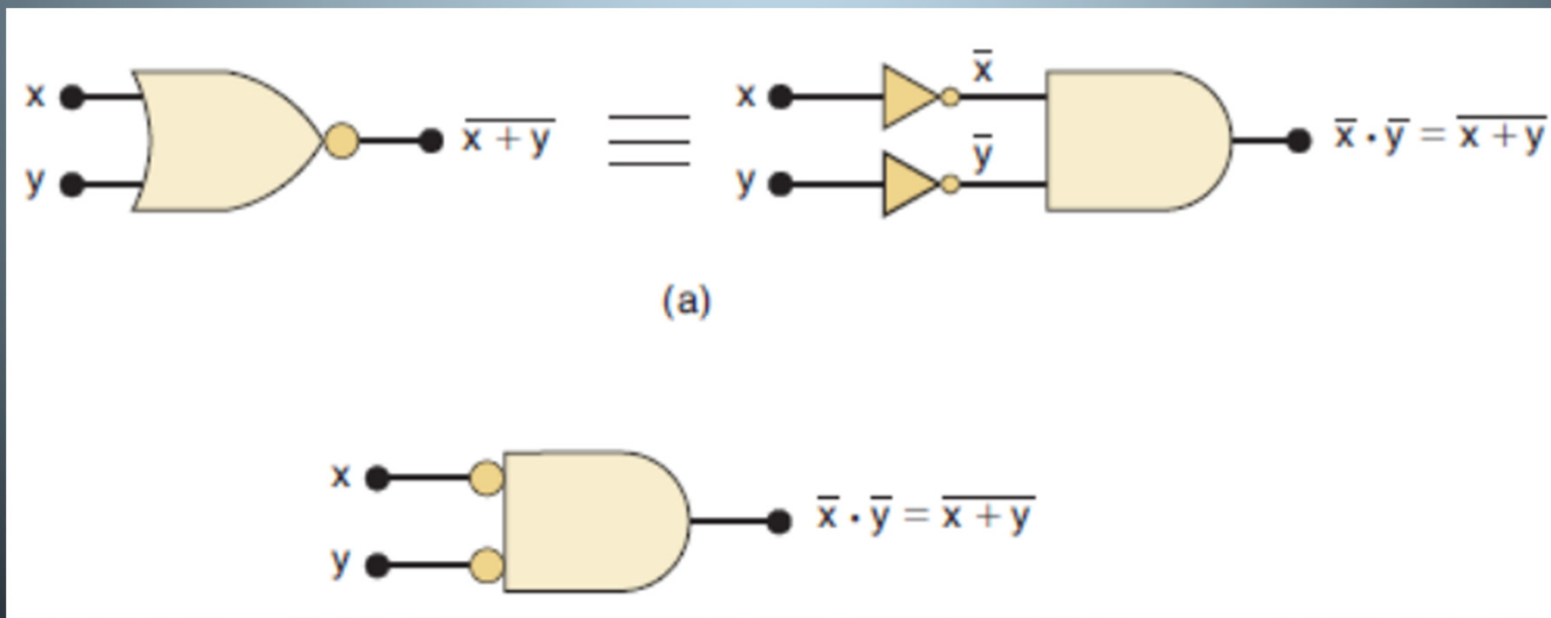
Theorem 1: $\overline{(\bar{X} + \bar{Y})} = \bar{X} \cdot \bar{Y}$

Theorem 2: $\overline{(\bar{X} \cdot \bar{Y})} = \bar{X} + \bar{Y}$

Inputs				Outputs of Theorem (1)		Outputs of Theorem (2)	
X	$\bar{X}$	Y	$\bar{Y}$	$\bar{X} + \bar{Y}$	$\bar{X} \cdot \bar{Y}$	$\bar{X} \cdot \bar{Y}$	$\bar{X} + \bar{Y}$
0	1	0	1	1	1	1	1
0	1	1	0	0	0	1	1
1	0	0	1	0	0	1	1
1	0	1	0	0	0	0	0

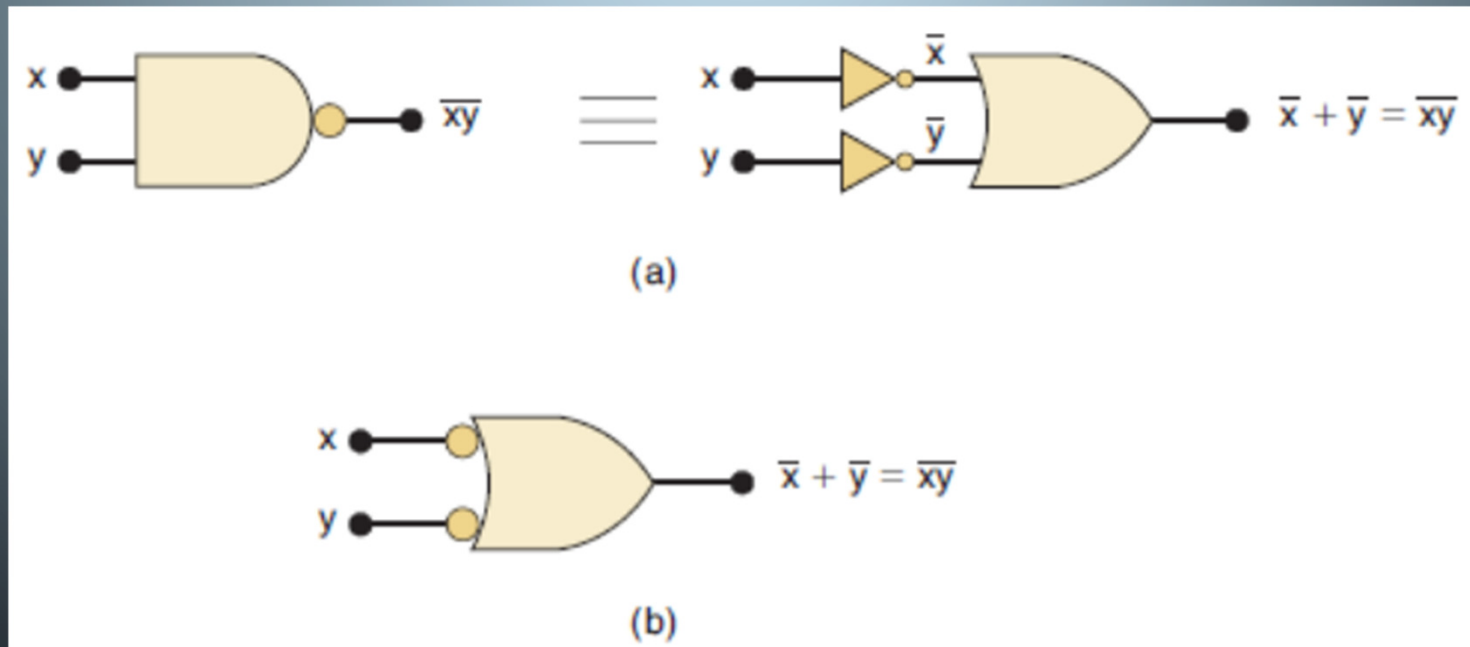
BOOLEAN ALGEBRA – DE-MORGAN'S THEOREM

Theorem 1: $\overline{(\bar{X} + \bar{Y})} = \bar{X} \cdot \bar{Y}$



BOOLEAN ALGEBRA – DE-MORGAN'S THEOREM

Theorem 2: $(\overline{X \cdot Y}) = \overline{X} + \overline{Y}$



EXERCISE

Simplify the expression $z = A\bar{B}\bar{C} + A\bar{B}C + ABC$.

Simplify $z = \bar{A}C(\bar{A}BD) + \bar{A}B\bar{C}\bar{D} + A\bar{B}C$.

Prove that $AB + BC + \bar{B}C = AB + C$.

Simplify the expression $AB + \bar{A}\bar{C} + A\bar{B}C(AB + C)$.

Simplify $Y = ABC + A\bar{B}C + AB\bar{C}$ to $Y = A(B + C)$.

Simplify the given Boolean expression $Y = \bar{A}\bar{B}\bar{C} + \bar{A}B\bar{C} + A\bar{B}\bar{C} + ABC$.

Simplify the expression $Y = \bar{A}C[\bar{A}BD] + \bar{A}B\bar{C}\bar{D} + A\bar{B}C$.

Prove the following Boolean expression $(A + B)(\bar{A}\bar{C} + C)(\bar{B} + AC) = \bar{A}\bar{B}$.

Prove that $\bar{A}\bar{B}\bar{C} + \bar{A}\bar{B}C + \bar{A}B\bar{C} + \bar{A}BC + A\bar{B}\bar{C} = \bar{A} + \bar{B} + \bar{C}$.

Simplify the expression $Y = (\bar{A}B + \bar{C})(\bar{A} + \bar{B} + C)$.



THANK YOU